

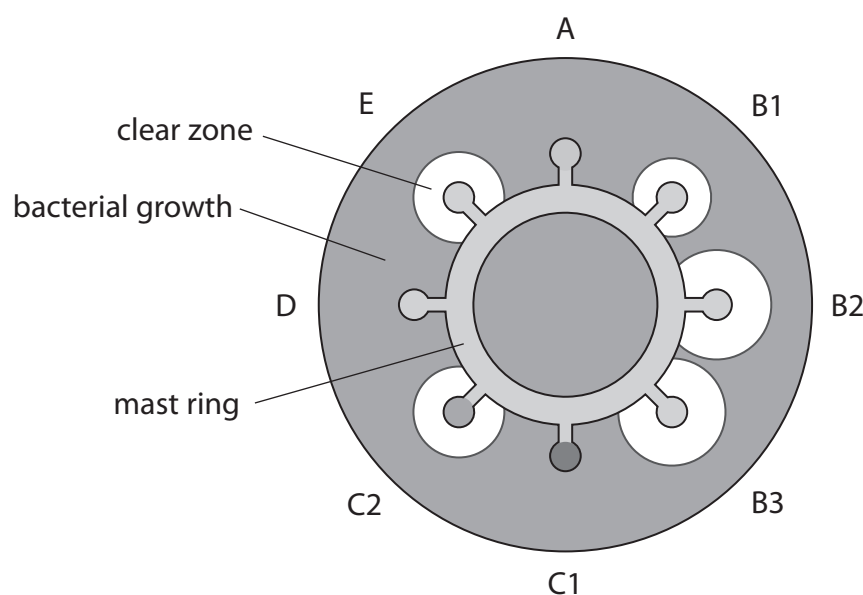
- 1 The resistance of bacteria to antibiotics can be studied using a mast ring.

A mast ring is made of filter paper.

Each part of the mast ring contains a different type of antibiotic or different mass of the same antibiotic.

The diagram shows a mast ring placed on agar in a Petri dish that has been spread with bacteria.

The table shows the mass of each antibiotic in the mast ring.



Part of mast ring	Antibiotic and its mass
A	100 μ g antibiotic A
B1	1 μ g antibiotic B
B2	10 μ g antibiotic B
B3	0.5 mg antibiotic B
C1	20 μ g antibiotic C
C2	200 μ g antibiotic C
D	100 μ g antibiotic D
E	100 μ g antibiotic E

Clear zones around the mast ring indicate that the growth of the bacteria has been inhibited.

The greater the diameter of the zone of inhibition, the more effective the antibiotic.

DO NOT WRITE IN THIS AREA

- (2)

- (1)

- ☐ **A** 0.5 times
- ☐ **B** 5.0 times
- ☐ **C** 50.0 times
- ☐ **D** 500.0 times

- (3)

[illegible]

(Total for Question 1 = 6 marks)

